

Excel Formulas

Lesson 25 — References, Operators and Copying

JKSSB Computer Awareness

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What a Formula Does

- ▶ A **formula** calculates a new value from other cells, from constants, or from functions.
- ▶ It is typed into the formula bar, and its result appears in the cell.
- ▶ Stored in the cell is the **formula**; shown on the sheet is the computed result.
- ▶ Excel recalculates the result automatically when a referenced cell changes.

The one idea

A formula is an instruction that tells Excel how to calculate the value of a cell.

A Formula Must Start with =

- ▶ Every formula begins with the equal sign =.
- ▶ =A1+A2 is calculated; A1+A2 entered without the = is stored as ordinary text.
- ▶ The same symbol also acts as a comparison operator that tests equality.

Trap alert (TRAP-30, PYQ-38)

A common wrong answer says a formula may start with +, %, or @. The exam rule is: a **formula must start with =**.

Formula Anatomy

- ▶ Take the formula `=A1*B1+100`.
- ▶ Leading `=`: starts the formula.
- ▶ Operands: cell references (A1, B1) and constants (100).
- ▶ Operators: `*` (multiply) and `+` (add).

Read it out

"Take the value in A1, multiply by B1, then add 100."

Arithmetic Operators

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
^	Exponent (power)
%	Percent

Exact symbols

Use * for multiplication and / for division; x and ÷ are not Excel operators. =2^3 gives 8.

Comparison Operators

Operator	Meaning
=	Equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
<>	Not equal to

A comparison formula returns TRUE or FALSE, not a number.

Text and Reference Operators

- ▶ `&` (concatenation) joins text: `= "Total " & A1`.
- ▶ `:` (range): `A1:A10` means all cells from A1 to A10.
- ▶ `,` (union) joins two ranges into one list.
- ▶ A space (intersection) gives the cells common to two ranges.

Why operators matter

An operator tells Excel what to do with the operands on either side of it.

Order of Evaluation (Precedence)

- ▶ Parentheses are evaluated first.
- ▶ Then percent, then exponent, then multiply/divide, then add/subtract.
- ▶ Then & (concatenation).
- ▶ Then the comparison operators.
- ▶ Use parentheses to force an order: $= (A1 + A2) * B1$.

Trap alert

$= 2 + 3 * 4$ is 14, not 20, because multiplication is done before addition.

Cell References Are Addresses

- ▶ A reference names a cell by its column letter and row number.
- ▶ A1 is column A, row 1; B5 is column B, row 5.
- ▶ A formula uses the **current value** of the referenced cell.
- ▶ Change the referenced cell and the formula result updates at once.

Relative References

- ▶ **A1** is a **relative** reference.
- ▶ It is stored as an offset from the formula's own cell.
- ▶ Copy the formula and the reference shifts by the same offset.

Example

=A1 placed in B1 and copied to B2 becomes =A2.

Absolute References

- ▶ **\$A\$1** is an **absolute** reference.
- ▶ The dollar sign locks **both the column and the row**.
- ▶ Copying the formula leaves **\$A\$1** unchanged.
- ▶ Use it for a constant such as a fixed rate or a fixed limit.

Remember

Absolute means anchored: it never moves when the formula is copied.

Mixed References

- ▶ **\$A1**: column locked, row relative.
- ▶ **A\$1**: row locked, column relative.
- ▶ The \$ locks whatever it directly precedes.
- ▶ \$ before a letter locks the column; \$ before a number locks the row.

Copying Formulas: What Changes

Reference	Locks	B1 copied to C2
A1	nothing	B2
\$A\$1	column and row	\$A\$1
\$A1	column	\$A2
A\$1	row	B\$1

Only the unlocked parts of a reference shift when the formula is copied.

Worked Example: A Fixed Tax Rate

- ▶ Prices are listed in column A; the tax rate is fixed in \$B\$1.
- ▶ Formula in C2: =A2*\$B\$1.
- ▶ Copy down — C3 is =A3*\$B\$1, C4 is =A4*\$B\$1; the rate stays \$B\$1.
- ▶ If the formula were =A2*B1, copying down would wrongly move the rate to B2, B3.

Common Formula Errors

Error	Meaning
#DIV/0!	Divide by zero or by an empty cell
#VALUE!	Wrong type of operand (text where a number is needed)
#REF!	A referenced cell was deleted, so the reference is invalid
#NAME?	Unknown function, or a misspelled name/text entry
#N/A	The value is not available (for example a failed lookup)
#NULL!	Two ranges that do not intersect

Trap Alert: References and Errors

- ▶ TRAP-30: a formula must start with = (PYQ-38).
- ▶ \$ before the letter locks the column; \$ before the number locks the row.
- ▶ Relative references change on copy; absolute references do not.
- ▶ #REF! means a broken reference; #DIV/0! means division by zero.

Lesson 25: Recall

- ▶ A formula begins with = and calculates a cell's value.
- ▶ Arithmetic operators: + - * / ^%; comparison operators include <> for "not equal".
- ▶ & joins text; : defines a range; , unions ranges.
- ▶ A1 is relative, \$A\$1 is absolute, \$A1 and A\$1 are mixed.
- ▶ On copy, only the unlocked parts of a reference change.
- ▶ Know the error codes: #DIV/0!, #VALUE!, #REF!, #NAME?, #N/A, #NULL!.